ENGINEERING PREVIEW Source checks pass. Cloud and physical acceptance remain open.

ThoxOS Ginux Desktop, shell and kernel proof

Latest image enhancements and four safety repairs, with authentic desktop captures and a fresh kernel lifecycle recording.

DOCUMENT / REVISION
THOX-GINUX-CONT-20261030 / 1.0

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CLASSIFICATION
Private engineering report

PRODUCT STAGE
Source continuation; no production qualification

01 Outcome and validation

The latest Debian 13 desktop/image work is integrated beside the existing Azure Compute kernel path. The first experience is clearer: choose the desktop, an authenticated browser shell, or the headless kernel lifecycle. Their patch sets, binaries and evidence remain separate.

172 passed
Image-source local CI; 1 font-path skip

62 passed
Root Azure/module/SSD; 27 subtests

53 passed
Independent targeted review; no skips

The 26 new regression cases are included in the 172 total. The independent 53-test run overlaps the implementation suite and is not additive. ShellCheck passed for all 24 script paths. These are source checks, not Docker, image-rebuild or Azure deployment passes.

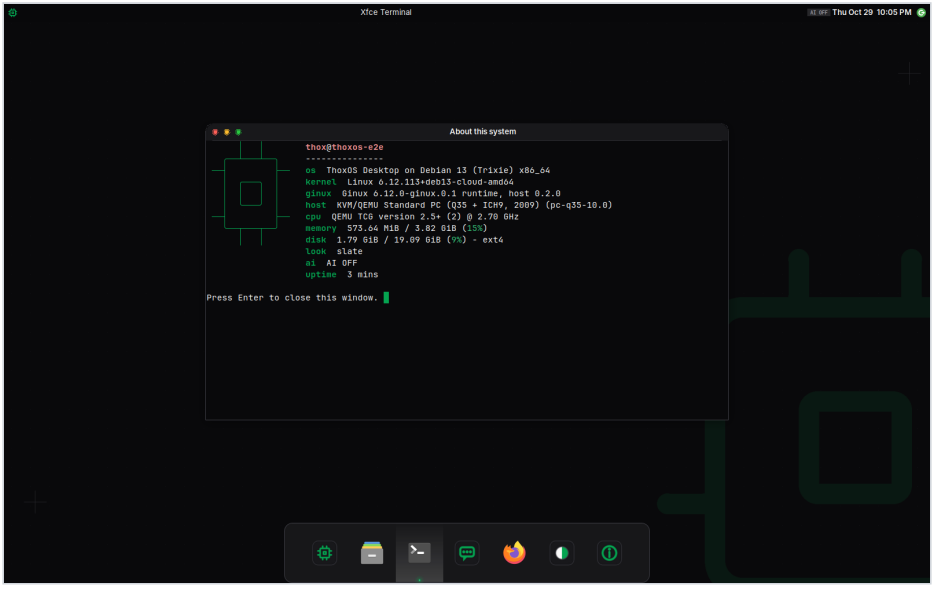
Four focused repairs

Area	Resulting behavior
Source preparation	Verify pinned source and patched bytes; refuse foreign, changed or stale checkouts instead of deleting them.
Docker configuration	Forward supported runtime settings through privilege drop using an explicit allowlist.
Hosted relay	Deny redirects so credential-bearing requests cannot move to an unintended destination.
Azure credentials	Remove the API key from customData; leave hosted AI off until deliberate root-owned configuration.

02 The desktop experience

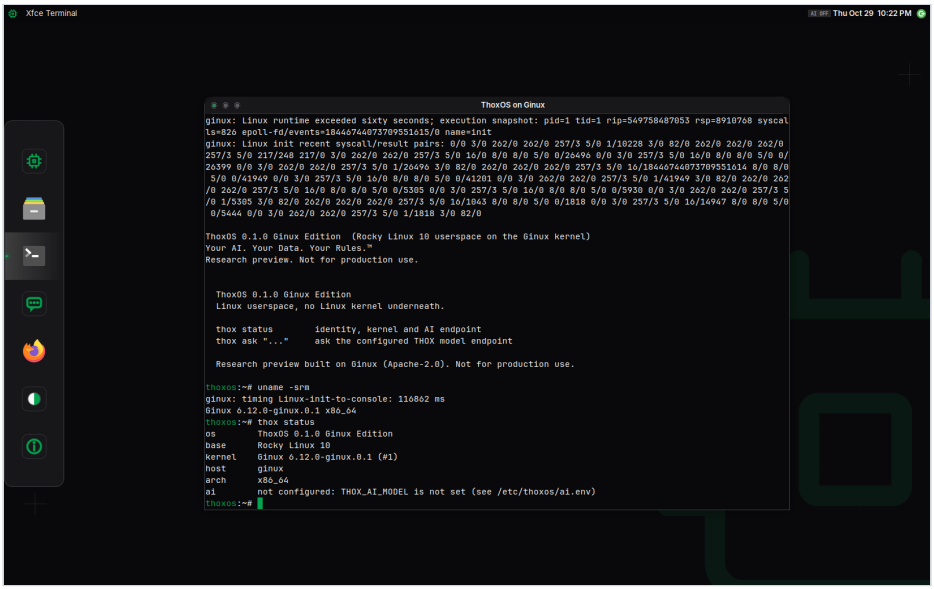
The image team's existing captures show a real ThoxOS Desktop on a Debian 13 host: TXF 5.2 looks, RDP, About, files and a Ginux console. The Ginux guest itself has no graphical desktop.

ACTUAL PRIOR IMAGE-TEAM CAPTURE



TXF Slate with About. Captured over RDP from the built VHD booted in QEMU software emulation on October 29. This is not an Azure deployment screenshot.

ACTUAL PRIOR IMAGE-TEAM CAPTURE

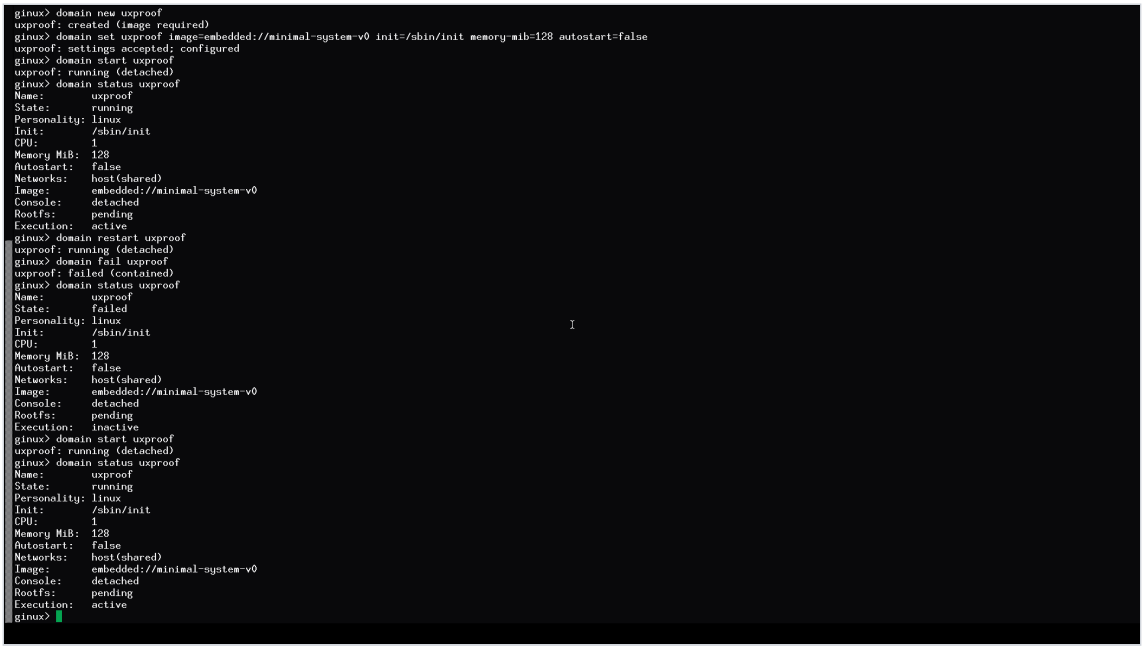


Console Pro with ThoxOS running in the user's Ginux VM. The prior desktop bench passed 23 checks; the headless bench passed 13; the SSD image USB-disk bench passed 6. These were virtualized image tests, not physical boot.

Reviewed source: image implementation `7f8f2007d3ae8da19ddb0309e99ee25904971cd1`. Existing image results are preserved evidence; the safety-repaired images have not been rebuilt in this continuation.

03 Fresh Ginux kernel proof

ACTUAL CURRENT QEMU SESSION



Fresh local QEMU TCG capture: the accepted root Azure EFI starts an embedded minimal-system domain, reports detached running state, restarts it, contains a failure and returns it to running. Automated operator inputs; 1 vCPU, 2 GiB RAM and OVMF. This proves the tested embedded workflow only.

Captured EFI SHA-256:
`3ebcc9043ef9dc8b7e75abc15b113a5810c8d39c0ad47143476a41caed24d5ba`

The recordings are real terminal framebuffer captures, not animated transcripts. The cold boot checks 11 operator commands and the recovered boot checks 4, with clean QEMU exits. They use the already accepted EFI on the same guest volume; no new kernel rebuild is claimed. Final hashes and execution details are in the continuation validation receipt.

Open the cold guest recording · Open the recovery recording · Repository experience note

Try the interactive experience

```
cd ginux-edition
make test
make demo-image
make selftest
make demo-shell
# In ThoxOS: uname -srm; thox version; thox status
```

Use the documented shell mode for interaction; the upstream systemd getty path does not complete password login. The browser shell binds to loopback and requires the generated credential. Each Docker or image build needs its own fresh acceptance.

04 Acceptance and next owners

Completed source work and remaining target gates

Area	Current evidence	Next acceptance / owner
Source repairs	Local CI, focused regressions and independent targeted review pass.	Coordinating session: bind final remote SHA, artifacts and readback in delivery receipt.
Docker and images	Prior image-team boots preserved. Current runtime denies mount/user namespaces; no Docker daemon or KVM.	Build/QA lane: rebuild patched images; rerun Docker, shell/ browser and VHD/SSD benches on a capable host.
Azure	No configured subscription/ identity; no cloud deployment, what-if or Gallery upload.	Azure owner: use the selected design, validate, deploy, inspect provisioning and measure nested KVM.
KnightHub	Connected devices offline; no physical SSD selected or written.	Hardware owner: read-only inventory, explicit target/action authorization, then physical boot and persistence.
MeshStack	No established MeshStack integration in this Ginux variant.	Existing Mesh owner: verify a live coordinator, nodes, workload and result before a distributed-inference claim.

The desktop VHD uses an unsigned loader and Standard security. The separate root Ubuntu marketplace-host design uses Trusted Launch. Neither is a Windows Azure Virtual Desktop session host. Hosted-model tests use local stubs; real hosted inference remains unrun.

Source and document provenance

- Existing main: `10787b1ae9dc024f069d229e83339876212afc5e`. Image input: `7f8f2007d3ae8da19ddb0309e99ee25904971cd1`, tree-equivalent to `98845311dd79f09fb96c25e7e6e9200ece7631ff`. Upstream Ginux: `36503bda2d87465765e1b94e65f9d100f27ed5be`.
- Image validation and raw evidence; root kernel validation. Historical results are not new execution.
- Microsoft customData guidance advises against sensitive values. The linked master-image guide concerns Windows AVD; this work uses Linux VMs with QEMU.

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